

AI-Driven Content Strategy Engine

Milestone 3 — Anticipated Q&A

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This document anticipates likely instructor and reviewer questions for the final presentation. Each question is paired with a concise answer that cites the specific result, figure, or section in the whitepaper. It serves as both a defense brief and a study guide for the live presentation.

Section A — Data and corpus

Q1. Why is the analysis run on a synthetic corpus and not on live Reddit data?

A. Three reasons. First, reproducibility: Reddit's content surface, ranking, and rate limits change without notice; if the grading run hits a 429 or a different week's top-100, the reported numbers shift. Second, time budget: 12,000 posts at the free-tier ceiling (100 RPM) is roughly 25 uninterrupted minutes per pull, which is fragile inside a single session. Third, methods validation: the corpus has known ground-truth effects baked in, so the analysis is judged on whether it recovers them — which is the right test for a methodology project. The repository ships a working RedditCollector, a trendspyg client, and a BlogScraper so a one-week replication on live data is a configuration change, not a code change.

Q2. How is the synthetic corpus 'realistic' rather than arbitrary?

A. Engagement is generated as the additive sum of literature-consistent topic, format, hook, day-of-week, daypart, and subreddit-size effects on the log scale, with normal noise. Subreddit-size is the largest single effect (a known confounder in Reddit research); topic is the largest content effect; effect sizes are small-to-small-medium, matching what published studies of organic engagement actually find. The distribution of scores is log-normal with the heavy right tail seen in real Reddit corpora (median 17, mean 26.4, max in the hundreds).

Q3. Why was pytrends swapped out for trendspyg?

A. pytrends was archived in April 2025 and is no longer maintained. trendspyg is the actively maintained drop-in successor that exposes the same interest-over-time, related-queries, and rising-queries calls. Milestone 1 was updated to reflect this once the source-recheck flagged it.

Q4. Where does the \$107B 2026 content-marketing figure come from?

A. Content Marketing Institute's 2026 B2B research, cited in the whitepaper references. The same source also reports that content is roughly 26% of total marketing spend in 2026 — both numbers are used in the deck and paper.

Section B — Modeling and statistics

Q5. OLS beat gradient boosting on R^2 . Is your nonlinear model broken?

A. No, that gap is expected on a corpus where the data-generating process is additive on the log scale. With only additive linear effects, a tree ensemble's extra capacity goes into modeling noise interactions and gives back a small amount of test-set R^2 . Both models agree on the predictor hierarchy (subreddit size → topic → hook → format → time), which is the actually-load-bearing finding. On live data with interaction effects (e.g., topic × subreddit), the gradient-boosted model is expected to overtake OLS — the pipeline keeps both.

Q6. Your η^2 values are small. Doesn't that mean the effects don't matter?

A. It means no single factor is a magic bullet, which is realistic for multifactorial content data. With 12 topics, 7 formats, 5 hooks, plus controls, you would not expect a single factor to dominate. The playbook is built on combined ordinal effects across multiple factors, not on any single factor's absolute magnitude — that's why the topic × format heatmap is the central visual and why the recommendation is to act on combinations (recovery + how-to + Tuesday morning), not isolated levers.

Q7. How do you defend the +37% generation uplift claim?

A. It is an explicit model prediction, not a measured outcome — and the document says so. The number is the mean difference in predicted log-engagement between sixty AI-conditioned drafts and sixty uniformly sampled baseline drafts ($n = 60$ each), bootstrapped 1,000 times to give a 95% CI of $[+0.137, +0.496]$ log units, which is +15% to +64% on the engagement scale. The interval excludes zero, so the conditioning is doing something inside the model. To turn predicted uplift into realized uplift requires a live A/B test on the platform; the recommended protocol is in the playbook section.

Q8. Why bootstrap instead of a t-test?

A. Two reasons: the predicted-engagement distribution is heavy-tailed even after the log transform, and the sample sizes ($n = 60$ vs. 60) are modest. Bootstrap CIs do not assume normality and are robust to the residual skew. A t-test would give a similar point estimate but a narrower (and overconfident) interval.

Section C — Leading indicators and Trends

Q9. RQ3 came back mixed. Should it have been dropped?

A. No — a per-topic answer is more useful than a categorical one. Google-Trends interest leads on-platform engagement on training (+2 weeks, $r = 0.26$) and gear (+5 weeks, $r = 0.11$) but not on recovery (−6 weeks) or sleep (−8 weeks). Reporting that honestly tells a client: 'use Trends as a leading signal on these topics, not those.' Dropping the analysis to keep a tidy

narrative would have been a worse decision under the rubric's ethics and limitations expectations.

Q10. How would lead-lag scale if you ran this on twelve niches instead of one?

A. Each niche-topic pair gets its own cross-correlation. The dashboard's lead-lag explorer page accepts (niche, topic) as a filter and surfaces per-topic best-lag and correlation, which is the deliverable form that scales.

Section D — Generation and LLM use

Q11. How is the generation step actually conditioned today vs. how it will be in production?

A. Today the M2 evaluation conditions by sampling categorical features from the empirically top-quartile buckets and scoring the resulting feature vector with the trained model. That tests whether the conditioning logic moves the model's prediction in the expected direction. Production conditioning is a structured LLM prompt that combines the same top combinations with the client's brand voice and asks for 20 drafts; the next-steps section in the M3 paper calls this out explicitly.

Q12. What stops the system from generating misleading or manipulative content?

A. Three layers. Scope: standard marketing topics in a defined niche; political persuasion, health claims, and financial advice are explicitly out of scope and are blocklisted in the generator's allowed-topic list. Disclosure: every generated draft is flagged as AI-assisted in the dashboard and the playbook, consistent with the May 2026 FTC guidance update. Sign-off: the dashboard has a human-review checkbox so no draft ships without a human review step.

Section E — Ethics, IP, and privacy

Q13. Reddit posts are public — why does aggregation still matter for ethics?

A. Even on public posts, users do not necessarily expect their submissions to be aggregated and modeled. The project's response is to analyze only at the aggregate / content-attribute level, drop usernames after a subreddit-level control is computed, and avoid any per-user behavioral modeling or demographic inference. That keeps the work inside the spirit of platform terms and inside what an IRB would call minimal-risk human-subjects research even though it is not formally human-subjects.

Q14. What about blog-article copyright when scraping?

A. The scraper stores titles, publish dates, word counts, declared topic tags, and (where exposed) share counts only — not full article bodies. That keeps the use inside fair-use bounds

for academic analysis and avoids any risk of redistributing copyrighted text. The generator's drafts are original model output, not paraphrase of any single source article.

Section F — Limitations and self-critique

Q15. If a reviewer says 'this is just descriptive statistics dressed up,' what do you say?

A. The project is more than descriptives, but reviewers should be able to see exactly where it goes further. RQ1 is a predictive model with held-out R^2 and SHAP-style importances. RQ2 is inferential — ANOVA with effect sizes. RQ3 is temporal — cross-correlation lead-lag detection. RQ4 is generative-evaluation — bootstrap CIs on a treated-vs-control prediction difference. Together those cover supervised learning, inferential statistics, time-series structure, and generative AI evaluation. That is the breadth-of-MSDS rubric criterion.

Q16. What would you do differently if you ran this again?

A. Three things. (1) Allocate the first week to a live one-week Reddit pull instead of synthesizing the corpus, so the M2 numbers are on real data. (2) Add a topic $\times$ subreddit interaction term to the OLS — the corpus only has main effects, but live data will not. (3) Replace the heuristic generator-conditioning step with a structured LLM prompt template earlier, so the +37% uplift is computed on LLM output rather than on feature samples that mimic LLM output.

Q17. If you had to defend the project in one sentence, what is it?

A. The project shows the full data-science arc — supervised learning, inferential statistics, time-series, and generative-AI evaluation — applied to a real and common business problem (content strategy), with limitations and ethics carried in the body of the work rather than in an appendix.